Paleomagnetic Indications of Megatectonic Movements in the Tethys

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Abstract. The paleomagnetic data obtained in northeast Italy, southern France, and northeast Spain differ *considerably and systematically* from those for meso-Europe. The deviation of the inclination of the paleomagnetic directions may be explained by assuming that the structural units of Italy, southern France, and Spain have moved westward with respect to meso-Europe. This late Paleozoic, Mesozoic, and Cenozoic westward drift of rigid blocks in the mobile Tethys zone is considered to be due to dextral shear movements, which developed in the Tethys zone during the northwestward drift of the Gondwana shields (Africa, Arabia, and India) and the contemporaneous eastward movement of meso-Europe. The deviation of the declination of the paleomagnetic directions in northeast Italy, Corsica, and northeast Spain are indicative of counterclockwise rotations. These rotations are thought to be related to the dextral shear movements mentioned above. They were caused by subsidiary left lateral shear which developed in the mobile zones bordering the primary dextral faults. Similar structural systems are found in the former Tethys areas of Turkey (Anatolia fault system) and northwest Africa (Atlas fault system). The main conclusion drawn from the paleomagnetic data is that the Tethys mobile belt was and still is a zone of primary dextral shear.

Introduction. Students of the University of Utrecht have investigated the paleomagnetism of samples from many parts of Europe. By courtesy of Professor J. Veldkamp, the collected samples were measured at the Geophysical Department of the Royal Netherlands Meteorological Institute. At first the studies were concentrated on the magnetic properties of Permian igneous rocks sampled in the more or less tectonically undisturbed areas of France, Germany, and Norway (see the publications of *Rutten et al.* [1957], *van Everdingen and Zijderveld* [1959], *van Everdingen* [1960], and *Nijenhuis* [1960]). In 1959, the study of the Permian rocks in tectonically disturbed areas was started and aberrant data were soon revealed by *Dietzel* [1960] and *van Hilten* [1960] for the southeastern Alps and by *van der Lingen* [1960] and *Schwarz* [1964] for the central Pyrenees. In 1960, 1961, and 1962, *de Boer* [1963] analyzed samples from the Vicentinian Alps (southwestern region of northeast Italy) as a part of a general geological study. The many well-exposed outcrops, and the abundant manifestations of igneous activity ranging in age from late Paleozoic to late Tertiary, render this region particularly suitable for a study of the paleomagnetic properties of Permian, Triassic, and Tertiary rocks. The data obtained in this area led to the new view concerning the megatectonics in the Tethys zone which is outlined in this paper.

The magnetic properties of the samples were determined with astatic magnetometers designed by As (for the technical details of these magnetometers the reader is referred to the publications by *As and Zijderveld* [1958] and *As* [1960]). After the initial measurements were made, the following corrections were applied.

A. Magnetic correction, i.e., a correction for the deviation caused by the unstable component resulting from a magnetization at a later, usually subrecent time.

B. Tectonic correction, i.e., a correction for the deviation caused by local deformations.

The magnetic correction is based on the studies of Zijderveld, van Everdingen, and As, who developed a method for separating the thermoremanent (stable) and viscous remanent (unstable) components of the magnetic vector by a progressive demagnetization of the oriented samples, i.e., by exposing them to an alternating magnetic field of stepwise increasing intensity. This method proved that the direction of magnetization of 80% of the samples changed con-

siderably during demagnetization, so that the magnetic correction is undoubtedly indispensable. The maximum intensities of demagnetization required to remove the unstable component varied considerably but were mostly higher than the generally used 150-oe value.

The local tectonic correction can only be applied to samples from regions which have not been deformed too much, i.e., from regions in which, during the deformations, no major internal (by plastic deformation) or external (by tilt) rotations and translations occurred This explains why most paleomagnetic studies are restricted to parts of the stable shields. In mountain ranges a detailed tectonic study is required in order to eliminate sampling areas in which rotations took place during tectogenesis. On the other hand, the tectonic correction can often be established by comparing the different directions of magnetization in rocks of the same age and composition but located in different areas [*de Boer,* 1963].

Paleomagnetic data. Table 1 summarizes the paleomagnetic data obtained in what is generally called meso-Europe (the part of Eurasia that has not been influenced by the Alpine orogenesis). Table 2 provides the paleomagnetic data for rock samples from different structural units in the Tethys zone (the part of Eurasia which was influenced by the Alpine orogenesis). The most reliable data for the Tethys zone are from rocks of Carboniferous, Permian, Triassic, and Tertiary age. Therefore an attempt will be made to compare these measurements with the measurements for contemporaneous rocks of meso-Europe.

The virtual magnetic poles for the different geologic periods in Russia, western Europe, and England are given in Table 3. The virtual magnetic poles of Russia, western Europe, and England during the late Paleozoic agree remarkably well, favoring the opinion that meso-Europe formed one stable block during this period. The distribution of the Permian and Carboniferous directions of remanent magnetization suggests a geocentric dipole field, which need not necessarily have been parallel to the earth's rotational axis. The axial dipole character of the earth's magnetic field has been demonstrated by *Hospers* [1954] for the late Tertiary, Quaternary, and historical times.

The polarity of the earth dipole changed sev-

TABLE 1. Paleomagnetic Data of Meso-Europe

Region*	Locality Lat.	Locality Long.	Comp.† of Specimen	Direction of Magnetization No.	Direction of Magnetization Decl. Incl.	Pole Position Lat.	Pole Position Long.	Fig. 1§	References
					Triassic				
WE	53°N	2°W	S	9	32/212 ± 26‡	43°N	133°E		Clegg et al. } *Cox and*
WE	55 N	5 W	S		214 − 48	54 N	118 E		Clegg et al. } *Doell*
WE	48 N	6 E	S	61	218 + 10	27 N	142 E		Clegg et al. } [1960]
EE	63 N	114 E	I		179 + 87	59 N	114 E		Komarov }
EE	66 N	88 E	I		90 + 71	48 N	148 E		Makarov
EE			I		67 + 84	61 N	117 E		Feinberg
EE	75 N	108 E	S	29	130 + 68	40 N	147 E		Gusev
EE	71 N	101 E	I	25	303 − 64	34 N	146 E		Gusev
EE	71 N	101 E	I	34	117 + 64	35 N	150 E		Gusev } *Kalash-*
EE	71 N	101 E	I	31	286 − 59	32 N	163 E		Gusev } *nikov*
EE	48 N	47 E	S	52	42 + 56	47 N	161 E		Khramov [1961]
EE	49 N	52 E	S	14	45/225 ± 46	49 N	158 E		Khramov
EE	53 N	52 E	S	9	220 − 51	54 N	164 E		Khramov
EE	48 N	38 E	S	26	39/219 ± 57	60 N	135 E		Khramov
EE	67 N	92 E	I	92	92 + 80	60 N	133 E		Kochegura }
					Permian				
WE	55 N	4 W	I	34	180 − 4	36 N	175 E	1	*Du Bois* [1957]
WE	55 N	4 W	S	26	187 − 6	37 N	163 E	1	*Du Bois* [1957]
WE	55 N	2 W	I	102	188 − 5	37 N	169 E	3	*Creer et al.* [1959]
WE	51 N	4 W	I	33	189 − 9	43 N	164 E	4	*Creer et al.* [1954]

TABLE 1. (Continued)

Region*	Locality Lat.	Locality Long.	Comp.† of Specimen	Direction of Magnetization No.	Decl.	Incl.	Pole Position Lat.	Pole Position Long.	Fig. 1§	References
WE	48 N	6 E	I	49	193	− 7	43 N	168 E	10	*Nairn* [1957]
WE	46 N	4 E	S	14	197	+ 6	38 N	162 E	9	*Nairn* [1957]
WE	43 N	3 E	S	53	189	+ 4	44 N	178 E	11	*Kruseman* [1962]
WE	43 N	3 E	S	92	198	− 6	48 N	164 E	11	*Kruseman* [1962]
WE	49 N	7 E	S	5	181	− 9	45 N	185 E	5	*Nairn* [1957]
WE	49 N	7 E	S + I	40	183	− 5	43 N	183 E	5	*Nairn* [1960]
WE	49 N	7 E	I	27	192	− 13	47 N	169 E	5	*Roche et al.* [1962]
WE	49 N	7 E	S + I	53	186	+ 4	38 N	179 E	5	*Nairn* [1960]
WE	49 N	8 E	S + I	33	182	− 7	45 N	185 E	5	*Nairn* [1960]
WE	48 N	8 E	I	18	157	+ 10	33 N	150 W	5	*Nairn* [1960]
WE	50 N	7 E	I	57	199	− 13	45 N	165 E	6	*Nijenhuis* [1960]
WE	60 N	10 E	I	484	204	− 36	47 N	147 E	2	*Van Everdingen* [1960]
EE	50 N	20 E	I	9	205	− 20	45 N	164 E	8	*Nairn* [1963]
EE	50 N	16 E	S + I	46	205	− 9	40 N	162 E	7	*Nairn* [1957]
EE	53 N	52 E	S	35	46/226	± 46	48 N	162 E	19	Khramov
EE	59 N	51 E	S	34	42/222	± 48	49 N	169 E	14	Khramov
EE	61 N	46 E	S	49	42/222	± 48	48 N	165 E	13	Khramov
EE	54 N	52 E	S	19	222	− 39	45 N	171 E	18	Khramov
EE	57 N	54 E	S	35	226	− 44	45 N	167 E	15	Khramov
EE	61 N	45 E	S	37	220	− 35	41 N	172 E	12	Khramov
EE	57 N	55 E	S	15	229	− 44	44 N	167 E	16	Khramov
EE	66 N	55 E	S	31	228	− 40	43 N	168 E	17	Khramov
EE	48 N	38 E	S	34	225	− 9	33 N	161 E	20	Kalashnikov
					Carboniferous					
WE	51 N	2 W	S	14	33	+ 35	48 N	126 E		Clegg et al.
WE	52 N	2 W	S + I	45	200	+ 15	27 N	156 E		Clegg et al.
WE	53 N	2 W	S	5	219	+ 41	6 N	142 E		Clegg et al.
WE	54 N	3 W	S	19	24	+ 23	44 N	142 E		Belshé
WE	54 N	3 W	S	18	27	+ 24	43 N	139 E		Belshé
WE	54 N	3 W	S	13	188	+ 9	30 N	167 E		Belshé
WE	53 N	1 W	S	103	27	+ 36	50 N	136 E		Belshé
WE	53 N	1 W	S	9	48	+ 47	47 N	105 E		Belshé
WE	56 N	3 W	I	63			18 N	154 E		Clegg et al.
WE	52 N	2 W	I	21	31	+ 3	32 N	139 E		Clegg et al.
WE	51 N	4 W	I	5	175	− 9	43 N	177 W		Blundell
EE	48 N	38 E	S	76	218	− 15	39 N	154 E		Khramov
EE	48 N	38 E	S	48	209	− 8	39 N	179 E		Khramov
EE	59 N	34 E	S	40	38/218	± 38	43 N	162 E		Khramov
EE	59 N	34 E	S	18	41/221	± 29	43 N	161 E		Khramov
EE	59 N	34 E	S	17	78/258	± 25‡	7 N	133 E		Khramov
EE	54 N	57 E	S	10	243	− 30	22 N	150 E		Khramov
EE	53 N	91 E	S		142	+ 87	48 N	98 E		Vlassov
EE	55 N	90 E	S		100	+ 50	20 N	153 E		Vlassov
EE	53 N	91 E	S		82	+ 78	49 N	129 E		Vlassov
EE	55 N	91 E	S		92	+ 70	41 N	141 E		Vlassov
EE	54 N	92 E	S		103	+ 45	12 N	156 E		Vlassov
EE	54 N	92 E	S		120	+ 84	46 N	110 E		Vlassov
EE	54 N	91 E	S		121	+ 84	46 N	110 E		Vlassov

* WE, western Europe; EE, eastern Europe.

† S, sediments; I, igneous rocks.

‡ 32/221 ± 26 = 32 + 26 (normal polarity) and 212 − 26 (reverse polarity).

§ The numbers in this column correspond to the numbers of the localities in Figure 1.

TABLE 2. Paleomagnetic Data of the Tethys

Age		Locality		Comp.* of Specimen	Direction of Magnetization, deg			Polarity†	References
					No.	Decl.	Incl.		
			Northeast Italy						
Tertiary	Middle Oligocene	Marostica	45°N 11°E	BE	16	184	− 61	R	*De Boer* [1963]
	Middle Eocene	Priabona	45 N 11 E	BE	29	353	+ 50	N	*De Boer* [1963]
Cretaceous		Schio	45 N 11 E	BS	4	356	+ 37‡	N	*De Boer* [1963]
Jurassic		Schio	45 N 11 E	BS	4	190	− 14‡	R	*De Boer* [1963]
Triassic	Norian	Schio	45 N 11 E	CS	3	15	+ 47	N	*De Boer* [1963]
	Ladino-Carnian	Tarvis	46 N 13 E	AI	28	28	+ 43	N	*Guicherit* [1964]
	Ladinian	Schio	45 N 11 E	BI	7	330	+ 49	N	*De Boer* [1963]
	Ladinian	Dobratsch	46 N 13 E	AI	8	18	+ 50	N	Lookeren (unpubl.)
	Ladinian	Fontanazzo	46 N 11 E	AI	4	26	+ 29§	N	*Guicherit* [1964]
	Anisian	Schio	45 N 11 E	AE	9	331	+ 49	N	*De Boer* [1963]
	Lower Anisian–Upper Scythian‖	Schio	45 N 11 E	AE	14	328	+ 40	N	*De Boer* [1963]
	Upper Scythian	Schio	45 N 11 E	AE	5	333	+ 40	N	*De Boer* [1963]
	Lower Scythian	Schio	45 N 11 E	AE	10	149	− 39	R	*De Boer* [1963]
	Lower Scythian	Schio	45 N 11 E	CS	5	149	− 35	R	*De Boer* [1963]
Permian	Late upper Permian	Staro ¶	45 N 11 E	AEI	5	330	+ 29	N	*De Boer* [1963]
	Late upper Permian	Camparmo ¶	45 N 11 E	AE	2	327	+ 37	N	*De Boer* [1963]
	Late upper Permian	Staro	45 N 11 E	AI	6	158	− 31	R	*De Boer* [1963]
	Late upper Permian	Camparmo	45 N 11 E	AE	4	151	− 29	R	*De Boer* [1963]
	Early upper Permian	Cortiana	45 N 11 E	CS	5	150	− 22	R	*De Boer* [1963]
	Late middle Permian	San Martino	46 N 11 E	CS	10	333	+ 26	N	*Guicherit* [1964]
	Late middle Permian	Coccau	46 N 13 E	CS	9	35	+ 24	N	*Guicherit* [1964]
	Early Permian	Chiusa	46 N 11 E	AE	8	173	− 8§	R	Nijenhuis (unpubl.)
	Early Permian	Lagorai	46 N 11 E	AE	38	137	− 8§	R	Findhammer (unpubl.)
	Early Permian	Merano	46 N 11 E	AE	51	164	− 10§	R	*Dietzel* [1960]
	Early Permian	Cavalese	46 N 11 E	AE	6	162	− 23	R	Lookeren (unpubl.)
	Early Permian**	Posina	45 N 11 E	AI	7	148	− 29	R	*De Boer* [1963]
	Early Permian	Val-di-Non	46 N 11 E	AE	39	150	− 31	R	*Van Hilten* [1960]
Carboniferous	Upper Carboniferous	Pramollo	46 N 11 E	CS	9	24	+ 40	N	*Guicherit* [1964]
			Northeast Spain						
Permian	Late Permian	Rio Aragon	42 N 3 E	AI	14	152	− 22	R	*Schwarz* [1964]
	Late Permian	Canfranc	42 N 3 E	AI	11	163	− 14	R	*Van der Lingen* [1960]
			Southern France						
Permian		Corsica	42 N 8 E	AE	15	141	+ 8§		*Nairn* [1963]
		Esterel	43 N 7 E	AE	500	207	− 22	R	Zijderveld (unpubl.)

* AE Acidic Extrusives; AI Acidic Intrusives; BE Basaltic Extrusives; BI Basaltic Intrusives; CS Clastic Sediments; BS Biochemic Sediments

† N Normal magnetization; R Reverse magnetization

‡ Unreliable; $\alpha 95 > 10°$.

§ Unreliable; no tectonic and/or magnetic correction applied.

‖ In part, the age of these rocks was determined by their paleomagnetic directions.

¶ Staro, Camparmo: In the Camparmo area the samples with normal polarity are younger than the ones with reverse polarity. The lava in the Staro area was magnetized in a normal field while the petrographically similar intrusives show both polarities. The age of the latter intrusive is in part derived from a paleomagnetic correlation.

** The age of these intrusive rocks was obtained by a paleomagnetic correlation.

Reliability: All data included in this table, with the exception of the ones marked above, have values for $\alpha 95$ between 4.2° and 11.5°. Because they do not have a Gaussian distribution and do not meet the fundamental requirements for the Fisher formula, the $\alpha 95$ values were not included in this table.

TABLE 3. The Virtual Magnetic Poles for Different Geologic Periods in Russia, Western Europe, and England

Period	Russia [*Kalashnikov*, 1961]	Western Europe [*van Hilten*, 1962]	England [*Creer et al.*, 1954]
Late Tertiary	74°N 183°E		89°N 5°E
Early Tertiary	64°N 160°E	80 5°N 161°E	74°N 133°E
Cretaceous		79°N 117°W	
Jurassic	54°N 159°E		
Triassic	49°N 144°E		43°N 131°E
Permian	44°N 166°E	43°N 169.5°E	43°N 164°E
Carboniferous	35°N 141°E	37.5°N 146.5°E	48°N 138°E
Devonian	31°N 158°E		30°N 159°E

eral times during geologic history. The Carboniferous magnetic pole was normal. Its polarity reversed in the early Permian and appears to have remained reversed for at least 50 m.y. The polarity changed again in the upper Permian (see data of Khramov in *Kalashnikov* [1961], *de Boer* [1963], and *Guicherit* [1964]), and after a short reversal during the lower Triassic (Scythian, see Table 2) it remained normal throughout the Triassic. Little is known about the Cretaceous and Jurassic. During the Tertiary a whole series of reversals seem to have taken place [*Glangeaud and Bobier*, 1963], indicating a considerable increase in the frequency of changes in the earth's magnetic polarity. The probable presence of a dipole field throughout geological times makes it possible to construct the equator and small circles of equal magnetic inclination (isoclines) for known magnetic poles; see *van Hilten* [1962]. The Carboniferous isocline map constructed for Eurasia shows that the inclination of the paleomagnetic direction in northeast Italy should have been about +7.5° (see Figure 2). Instead, +40° has been measured (see Table 2), a difference of 32.5°. The comparison of the Permian data with the expected values is even more interesting (see Figure 1). The measured inclinations of the paleomagnetic samples from northeast Italy, southern France, and northeast Spain differ by 26°, 26°, 25°, respectively, from the expected inclinations. A paleomagnetic inclination of +25° would be expected for the Triassic in northeast Italy (see Figure 2); instead, inclinations ranging from +40° to +50° (see Table 2) were measured, an average difference of 20°.

It may be concluded that the Carboniferous, Permian, and Triassic inclinations differ systematically from those expected if the calculations are based on meso-European magnetic poles.

Review of explanations for deviating magnetic directions. Several factors may have caused the differences in the paleomagnetic data obtained from contemporaneous rocks sampled in different areas. The most common and most important factors are (*a*) instability of magnetization, (*b*) incorrect or incomplete tectonic correction, (*c*) particular composition of the magnetic minerals, (*d*) metamorphism and/or magnetostriction of the magnetic minerals, (*e*) secular variation, and (*f*) megatectonic translations and/or rotations of the sampling areas.

To determine the most probable cause of the deviations in the Alpine areas, we consider the data obtained by *de Boer* [1963] in the Vicentinian Alps (northeast Italy). As all samples were progressively demagnetized (up to 900 oe), major errors due to the instability of magnetization (*a*) have probably been avoided. Also, important errors in applying the tectonic correction (*b*) have not been made, since, before sampling, relatively undisturbed outcrops were selected on the basis of detailed tectonic studies. The paleomagnetic directions have been obtained from acidic and basic igneous rocks and from sediments. The different chemical compositions of samples of the same age and with about the same direction of magnetization (see Table 2) make it unlikely that their deviation is the result of a particular Fe/Ti ratio or other mineralogical property (*c*). The influence of meta-

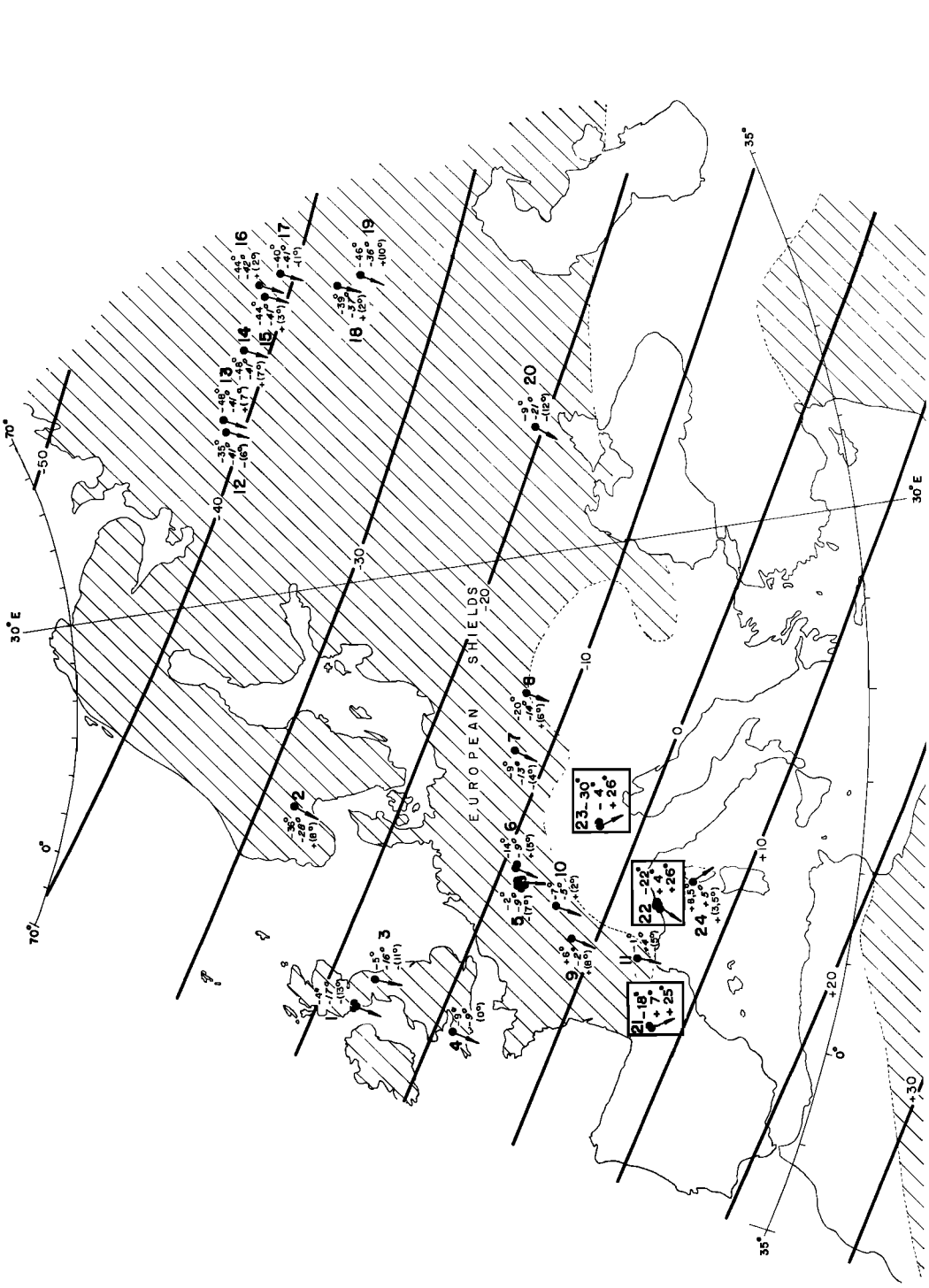


Fig. 1. Magnetic isocline map of the Permian. Base of arrow: location of sampling area. Arrow: direction of magnetic declination. Top number: measured magnetic inclination. Middle number: expected magnetic inclination. Bottom number: difference between the measured and expected magnetic inclinations.

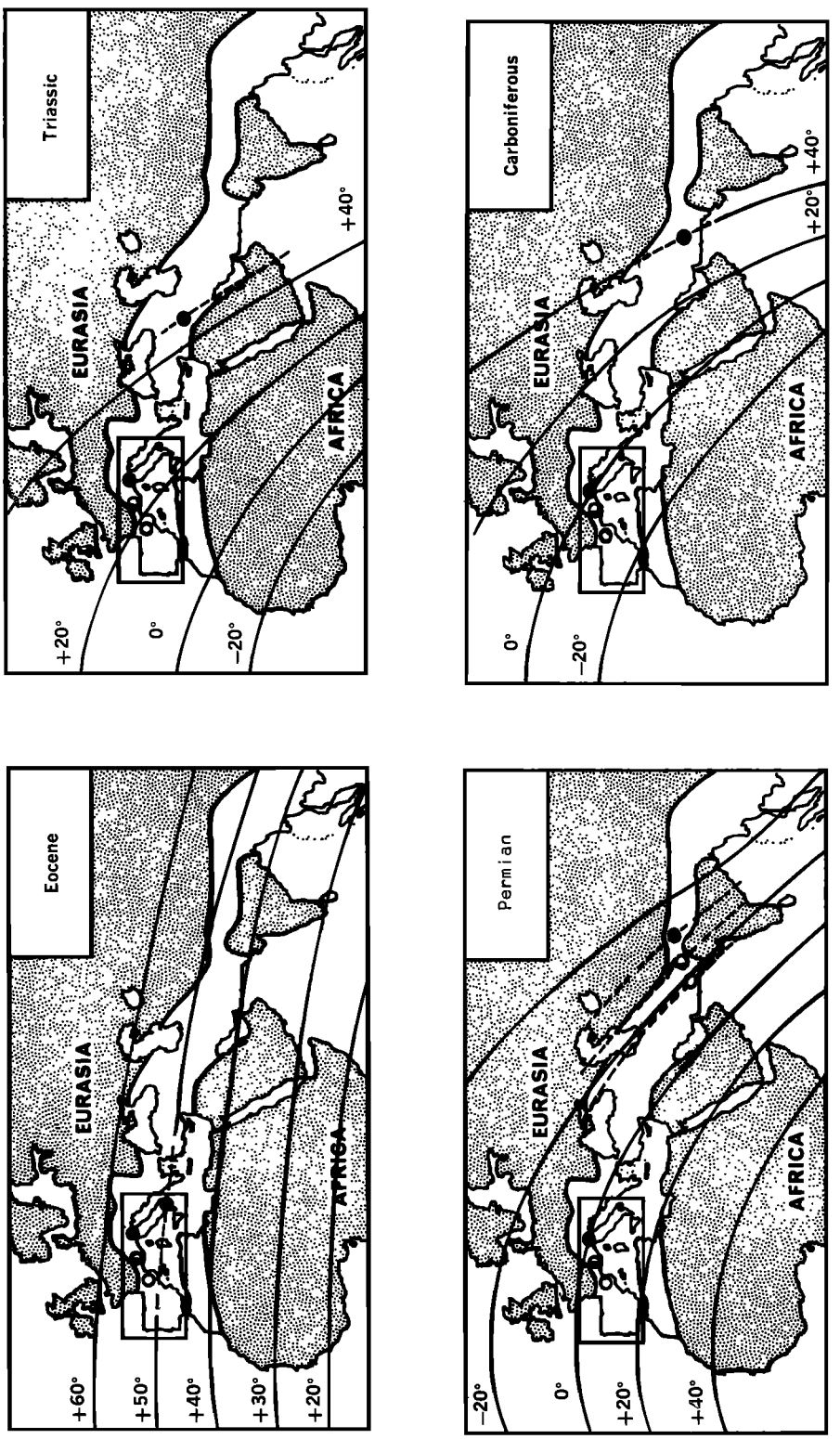


Fig. 2. Present-day map showing the European late Carboniferous, Permian, Triassic, and Eocene isoclines. Average European poles used: Carboniferous, 35°N, 141°E [*Kalashnikov*, 1961]. Permian, 43°N, 169°E [*van Hilten*, 1962]. Triassic, 49°N, 144°E [*Kalashnikov*, 1961]. Eocene, 80°N, 161°E [*van Hilten*, 1962]. Earlier locations of northeast Italy, southern France and northeast Spain are indicated respectively by filled circles, half-filled circles, and open circles.

morphism and magnetostriction (*d*) might be expected in an acidic igneous unit (Guizza-Faedo Mountains, north of Schio, Vicentinian Alps). This unit is at the northern border of the Schio flexure-fault, which caused considerable drag, having a throw of about 1½ km. During the Triassic, the flexure-fault served as a pathway for ascending hydrothermal solutions which locally changed the mineralogical composition of the sampled igneous rocks. Nevertheless, the paleomagnetic directions measured in the samples taken from this unit are not notably different from those obtained in other rocks of the same age, type, and composition (Alba-Fratte and Bafelant-Cornetto units). It may be concluded that the influence of the pneumatolitic-hydrothermal metamorphism and magnetostriction by tectonic deformation is below the critical value in the Vicentinian Alps, and it seems that the same can be said for all of northeast Italy, northeast Spain, and southern France. In the Vicentinian Alps the deviation of the paleomagnetic directions with respect to the European ones appears to have been constant during the whole Permo-Triassic. As secular variations (*e*) mainly occurred in relatively short periods, and as our data represent the mean values over very long periods, these variations cannot have caused the deviating directions. Therefore the last-mentioned possibility, namely, that large megatectonic displacements and/or rotations of the sampling area (*f*) took place, seems to be the most probable explanation of the observed paleomagnetic deviations between the Alpine mountain system and the more stable Eurasian shields.

Data based on the deviating inclinations. Two conclusions can be drawn from the magnetic inclinations obtained for northeast Italy (southern France, and northeast Spain). See Table 2.

A. The inclination of the paleomagnetic directions gradually increased during the late Permian and Triassic (from 30° to 50°) after a decrease in late Carboniferous and early Permian time (40° to 30°). The inclination may have been constant during the Jurassic and Cretaceous (±50°), but it changed abruptly in post-middle Eocene and pre-middle Oligocene time (from 50° to 61°).

B. The inclination of the magnetic vectors in the sediments is usually smaller (up to 25%) than in contemporaneous igneous rocks. This may be explained by a slight rotation of the magnetic sedimentary particles during sedimentation and/or compaction.

The Carboniferous magnetic pole of Russia (Table 3) will be taken as representative for the meso-European pole. Application of *van Hilten's* [1962] paleo-isocline (lines of equal magnetic inclination) method establishes the Carboniferous magnetic field of meso-Europe very well (Figure 2). The inclination measured in the Carboniferous rocks of northeast Italy averages +40°, which means that during this period this region must have been on or near the +40° isocline. This small circle on the northern hemisphere locates the geographical latitude of northeast Italy with respect to the Carboniferous pole. It is not possible to determine the geographic longitude of northeast Italy by paleomagnetic directions; for this purpose we refer to additional geological and paleoclimatological data. Because the crustal unit of northeast Italy was located in the Tethys zone and can never have occupied a position on the older meso-European shields, its location (relative to meso-Europe) is indicated by that part of the appropriate isocline which coincides with the contemporaneous Tethys belt. During the Carboniferous, northeast Italy must have been located (relative to meso-Europe) on the intersection of the +40° isocline and the Carboniferous Tethys zone. Paleomagnetic data indicate that the main axis of the late Paleozoic Tethys coincided more or less with part of the present equator. The northern border of the Tethys in Asia had about the same location as the present Himalaya mountain range. Thus it may be concluded that during the Carboniferous northeast Italy was located somewhere near the intersection of the +40° Carboniferous isocline with the present Himalaya Mountains (see Figure 2).

The Permian magnetic poles of *van Hilten* [1962] (western Europe) and *Kalashnikov* [1961] (eastern Europe) are almost identical. Both can be considered representative for the Permian pole of meso-Europe. According to *van Hilten* [1960] and *de Boer* [1963], the average inclination of the Permian igneous rocks of northeast Italy is −30° (based on 68 samples). The relatively low inclination of the Permian magnetic directions measured by *Dietzel* [1960] (Table 2) is thought to be due to the measuring of altered samples, which could not be

completely cleaned of additional magnetic components [*van Hilten,* 1960]. Nijenhuis (unpubl.) and Findhammer (unpubl.) (Table 2) could not apply a tectonic correction, which seems to account for the low inclinations. Their data are accordingly considered to be unreliable. The reason for using only the paleomagnetic data for the igneous rocks to obtain an indicative inclination is that this inclination is frequently lower in sediments than in contemporaneous igneous rocks (average inclination of Permian sediments in northeast Italy is −24°). The average Permian inclination in the igneous rocks from southern France is −22° [*Zijderveld,* 1964; 500 samples]. This value is one of the most reliable for western Europe. The average Permian inclination of the igneous rocks from northeast Spain is −18° [*Schwarz,* 1964; *van der Lingen,* 1960; 25 samples]. In their present positions these regions should have had inclinations of −4°, +4°, and +7°, respectively (see Figure 1). It can be concluded that northeast Italy, southern France, and northeast Spain not only show the same large systematic deviation in inclination (+26°, +26°, and +25°, respectively) but also have measured values which diminish from east to west at about the same rate as the expected values.

Eurasia	Permian magnetic pole Triassic magnetic pole	43°N, 169.5°E 52°N, 154°E	*van Hilten* [1962]
Northeast Italy	Permian magnetic pole Triassic magnetic pole	51°N, 120°W 58°N, 112°W	*de Boer* [1963]

As these *systematic and constant* deviations are restricted to the northern margin of the former mobile Tethys zone (at present, little is known about paleomagnetism of its southern margin), megatectonic movements of large parts of the Tethys appear to have played an important role. In the same way as was done for the Carboniferous, it may be concluded that during the early Permian northeast Italy was located somewhere near the intersection of the Permian −30° isocline with the present Himalaya Mountains (see Figure 2). The paths of the Permo-Triassic poles of Eurasia and northeast Italy are similar distances apart, though they differ considerably in location.

This coincidence can be used to get a more precise indication of the Permian longitude of northeast Italy. As the Permian and Triassic poles of northeast Italy probably coincided with the contemporaneous meso-European poles, northeast Italy's Permian location (with respect to the present location of meso-Europe) can be established at 36°N, 74°E, which is in the westernmost part of the Himalaya Mountains as predicted by the intersection of the −30° isocline and the northern border of the Permian Tethys zone. Because of the close relationship (the same *systematic and constant* deviations of their Permian magnetic vectors) between the data obtained in northeast Italy, southern France, and northeast Spain, we can conclude that the distances between crustal blocks did not change significantly during their post-Permian movements. Because of the lack of Triassic data, this cannot be established with more certainty.

From the Triassic magnetic isocline map of Eurasia, it follows that northeast Italy should have been situated on or near the +27.5° isocline during the Triassic, if it was in its present location. The measured inclinations, however, range from 40° to 50° (increasing from old to

TABLE 4. The Similarities between the Expected and the Measured Permian Inclinations of the Different Structural Units in the Tethys

	Northeast Italy	Difference	Southern France	Difference	Northeast Spain
Average measured inclination	−30°	8°	−22°	4°	−18°
Expected inclination	−4°	8°	+4°	3°	+7°

young). This favors the opinion that during the Triassic northeast Italy was located somewhere in what is now Iran (see Figure 2).

The position of the Eocene magnetic pole has been established by *van Hilten* [1962] at 80°N, 161°E, on the basis of data published by *Cox and Doell* [1960]. In its present location northeast Italy should have had an inclination of about +55° (see Figure 2) during the Eocene. However, mostly lower inclinations are found in the Vicentinian Alps. The average measured inclination is +50° ($\alpha 95 = 5.8°$). The difference between the measured and calculated inclinations accordingly is 5°, which falls within the limits of the errors that can be made during measurement. However, during the middle Oligocene the inclination of the magnetic field had reached about its present value (−61°; $\alpha 95 = 6.0°$; see Table 2). During a relatively short period (25 m.y.), therefore, the inclination increased considerably. As a southward extension of the Alpine sedimentary basins is generally accepted, and as the main phase of the Alpine deformations occurred after the middle Eocene, a northward movement of northeast Italy seems quite probable. The intersection of the Eocene +50° isocline and the Tethys zone cannot be accurately determined because the isocline cuts the northern margin of this zone at a small angle. The point of intersection accordingly may shift from east to west between Italy and Turkey. The location on the +50° isocline which is nearest to the present position of northeast Italy is in the southernmost part of Italy (about 40°N, 15°E). If northeast Italy had been situated here during the Eocene, it must have moved northward over about 800 km, a distance which falls well within the limits determined by most Alpine geologists.

Data based on the deviating Permian declinations. The declinations of the Permian and Triassic directions of magnetism in meso-Europe and in the Tethys zone differ significantly. The declinations of the magnetic directions obtained from meso-European rocks point roughly southwestward; the declinations measured in rocks sampled in northeast Italy, Corsica, and northeast Spain have a southeast direction. The deviations of the Permian magnetic declinations in the Tethys zone cannot be attributed to irregular courses of the isogones (lines of equal declination), as no distinct irregularities are indicated by the Permian magnetic data of meso-Europe. Moreover, these deviations are restricted to the Cenozoic orogenic belt, in which important rotations and translations of the structural units have taken place.

Nonlatitudinal translations of crustal units will cause changes in the values of the paleomagnetic declinations. The amount of this change has to be known before the rotations of the crustal units around an axis at right angles to their surface can be determined.

If northeast Italy was located at 30°N, 72°E, during the Permian, a westward translation over 54° is necessary to let this crustal unit occupy its present location at 45°N, 11°E. This translation over the surface of our earth equals a rotation of 54° around an axis with poles at 69°N, 170°E, and 69°S, 170°W.

The Permian magnetic pole of meso-Europe was located at 42.5°N, 169.5°E. The declination in northeast Italy during the Permian accordingly was 230°. The Permian declination in the region presently occupied by northeast Italy amounted to 196°. When northeast Italy moved westward (rotated around the axis with a pole at 170°E, 69°N), its Permian declination decreased by 15° and became 215°; however, a declination of 150° was measured. The difference of 65° accordingly must be due to a rotation of the crustal unit involved around a subvertical axis. This rotation has been anticlockwise. Northeast Spain also appears to have rotated anticlockwise, the angle being closer to 50°. Thus there is a significant resemblance between the deviations of the declinations as well as the deviations of the inclinations of the paleomagnetic directions in northeast Italy and northeast Spain. The samples from Corsica, which have not been magnetically cleaned, indicate a similar large anticlockwise rotation. The deviating declination of southern France (207°) indicates a westward translation. This crustal unit, however, did not rotate around a subvertical axis.

The rotations probably originated during the relative westward drift of the crustal blocks of Italy and Spain (and Corsica) in the mobile Tethys zone. A hypothesis concerning these anticlockwise rotations, based on geotectonic features, has been expressed by *Carey* [1958], who is of the opinion that the rotations are caused by sinistral shear.

In summary, the following conclusions can be deduced from the paleomagnetic data of northeast Italy:

A. The differences between the inclinations of the magnetic directions in contemporaneous rocks of meso-Europe and northeast Italy decreased from 32.5° in the late Carboniferous through 26° in the Permian and 20° in the Triassic to 5° in the Eocene, suggesting a westward drift of northeast Italy with respect to meso-Europe. The effect of this westward drift on the locations of the historic magnetic poles was superimposed on the general pattern of polar wandering.

B. The probable locations occupied by northeast Italy with respect to meso-Europe (provided that this continent stayed fixed) are Carboniferous, 30°N, 60°E; Permian, 30°N, 72°E; Triassic, 35°N, 40°E; Eocene, 40°N, 15°E; Present, 45°N, 11°E.

C. The velocity of this westward drift (provided that meso-Europe stayed fixed) ranges from 2.5 to 4 cm/yr, which is in accord with the estimated velocities of convection currents (2.5 to 5 cm/yr).

D. The differences between the declinations of the magnetic directions in contemporaneous rocks of meso-Europe and northeast Italy suggest an anticlockwise rotation (65°) of the latter structural unit with respect to the meso-European shield.

E. The westward drift of northeast Italy is thought to have started in the Permian; the rotation of this structural unit took place in late Triassic time.

As the Permian data indicate a close relationship between the structural units of northeast Italy, southern France, and northeast Spain, these regions are thought to have more or less the same structural history.

Interpretation of the dominant structural deformation in the Tethys zone. The relative westward drift of the crustal units of Italy, Spain, and southern France (and Corsica) with respect to Eurasia, as indicated by the paleomagnetic data, is thought to have been caused by the relative displacement of the Gondwana shields with respect to meso-Europe. These crustal units are thought to have been the northern outposts of the African continent. The Gondwana shields moved northwestward, the meso-European shields eastward, causing the formation of a dextral shear zone (the Tethys). The dextral movements which started in Permian time were interrupted during the Cenozoic when the African and meso-European shields approached each other to such a degree that the potential energy in the compressed Tethys could no longer be released by tangential movements but accumulated and caused epeirogenetic deformations ultimately resulting in the Alpine orogeny. In late Cenozoic and present times the lateral movements continued, as is indicated by recent tectonic and seismic activities.,

The dextral character of the megatectonic shear movements in the Tethys zone is determined from the following:

A. The presence and shape of the narrow belt of negative gravity anomalies in the eastern Mediterranean region. This belt of negative anomalies, which is still subsiding, cannot be considered to be due to an isostatic compensation but represents a tensional phenomenon. Its S-shape is the form to be expected in a dextral shear zone.

B. The seismic activity in the Mediterranean area reveals that most earthquakes originate on wrench fault planes. Of the earthquakes considered by *Scheidegger* [1963], 88.2% were the strike-slip type. The fault plane solutions for earthquakes in the Mediterranean region [*Hodgson and Cock,* 1956, 1957; *Hodgson and Stevens,* 1958; see *Aki,* 1964] reveal two fault types: approximately E–W directed dextral strike-slip faults and approximately NW–SE directed normal faults with sinistral transcurrency. Together, these faults form a system which could be produced by a WNW–ESE tangential compression.

C. A comparison of the paleomagnetic polar wandering curves of Africa, Europe, and North America led *Nairn* [1964] to conclude that 'the curves suggest a large, predominantly latitudinal, post-lower Jurassic/pre-Miocene movement of Africa with respect to Europe and North America.' From the figures it can be deduced that these movements must have been dextral. The paleomagnetic data of the Tethys zone indicate that the shift started earlier than in early Jurassic time, namely in the Permian.

D. The tectonics in the Mediterranean region reveal a dominance of dextral shear faults. The most important are the north Anatolia wrench fault system in Turkey and the southern Atlas

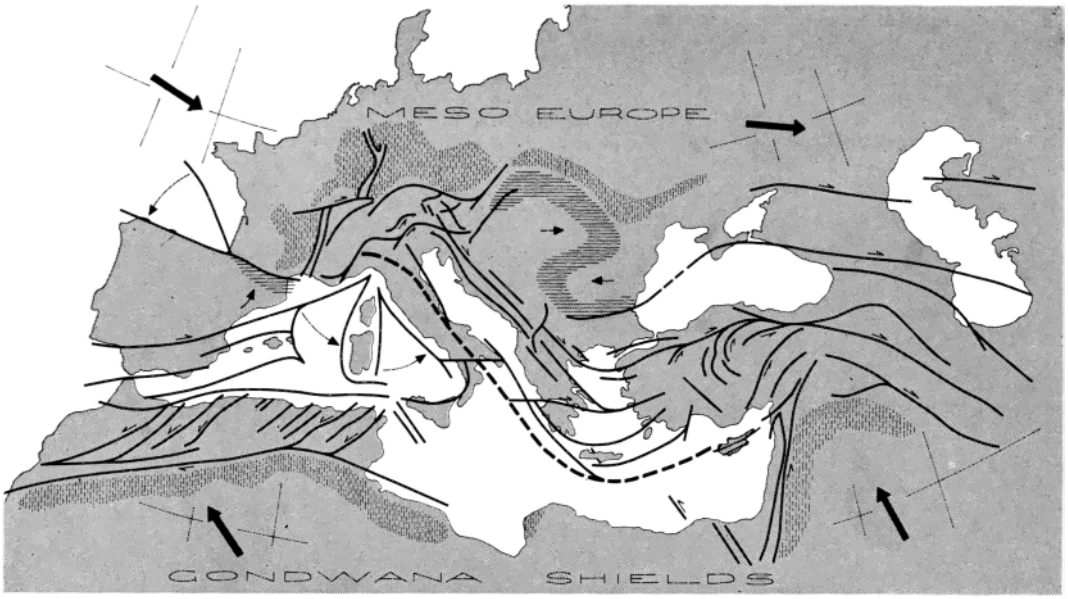


Fig. 3. Structural sketch map of the western Tethys. Solid line: main wrench fault. Broken line: axis of the belt of negative gravity anomalies. Small arrows: rotation of smaller units. Large arrows: direction of drift of the main units.

wrench fault system in northwest Africa. The northern Anatolian dextral wrench fault system extends for more than 3000 km from eastern Greece through Turkey and Persia and into the Persian Gulf [*Pavoni,* 1961]. Field data for Turkey indicate a cumulative offset of 400 km. In one cycle of seismic and tectonic activity (1939–1953) the total throw amounted to 1 meter (northern block down) and the total offset 2 to 3 meters (northern block eastward). The southern Atlas dextral wrench fault system can be followed from the Canary Islands (offshore Rio de Oro) along the northern border of the African shield [*Rod,* 1962] into the Gulf of Gabes (Tunisia). According to *Furon* [1963] it continues through the eastern Mediterranean into Turkey. Its cumulative offset is thought to be about 160 km (southern block westward).

Other important E–W dextral wrench faults are the Moroccan fault [*Carey,* 1958] which runs along the north coast of Morocco, Algeria, and Tunisia, the Trikkeri-Canal fault which sets off the northern part of the Greek seismic arc, and the faults in the Caucasus area (see Figure 4). These dextral wrench faults occur en echelon throughout the Alpine mountain system from the Atlantic to the Pacific [*Pavoni,* 1961], and they seem to represent one big dextral shear fault zone which forms part of a still bigger worldwide meridional shear system. This hypothesis concerning the dominance of dextral shear in the Tethys belt is in contrast with the sinistral shear supposed by *Carey* [1958]. The reason for this difference is in the counterclockwise rotations of the structural units of Spain, Corsica (and Sardinia), and Italy, which according to Carey are due to primary sinistral shear and which in my opinion are caused by second-order sinistral shear which developed in a primary dextral shear zone.

E. Structural analyses of the northwestern part of the African craton reveal a dominating NNW–SSE stress which resulted in an ENE–WSW megafolding; Tindouf syncline, Yetti Eglab anticline, Taoudenni syncline, Liberia–Upper Volta anticline [*Sougy,* 1962]. The Atlas mountain chain also indicates a NW–directed stress. According to *de Sitter* [1956] the post-Eocene stress had a NW direction and the post-Miocene stress had a NNW direction. Structural data indicating a NW drift of Arabia have been published by *Swartz and Arden* [1960]. The almost uniformly directed NW strain patterns of Africa and Arabia probably resulted from the northwestward drift of these fragments of the former Gondwana continent.

Assuming that the crustal units in the Tethys zone have drifted westward over a considerable distance, we should expect differences in the fossil content of the north and south Alpine deposits. These differences will be only slight for Cenozoic time, since the distances between these crustal blocks and the meso-European continent were greatest in late Paleozoic time and decreased considerably during the Mesozoic and Cenozoic. Also, the worldwide transgressions and uniform sedimentation in late Mesozoic time did not favor important faunal differences. The best evidence, accordingly, must come from the Permo-Triassic data. It has long been known that the upper Permian Bellerophon dolomites of the southern Alps cannot be correlated with the Zechstein evaporites north of the Alps. They are thought to have been deposited in basins separated by very long and narrow ridges. The faunal association in the Bellerophon dolomites, however, resembles very much the faunal association in the Indian Productus limestones of the Himalayas [*Brinkmann,* 1960]. Also the older cephalopod zones of the Scythian (lower southern Alpine, Triassic) are known only in the Himalayas and Armenia. It is difficult to compare these deposits with the lower Triassic of Germany since the latter are predominantly continental. The westward drift provides an explanation for the presence of the same association of cephalopods in contemporaneous sediments deposited in relatively narrow and shallow geosynclines, several thousand miles apart.

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References

Aki, K., Study of Love and Rayleigh waves from earthquakes with fault plane solutions or with known faulting, *Bull. Seismol. Soc. Am.,* in press, 1964.

As, J. A., Instruments and measuring methods in paleomagnetic research, *Mededel. en Verhandel. Koninkl. Ned. Meteorol. Inst.,* no. 78, 1960.

As, J. A., and J. D. A. Zijderveld, Magnetic cleaning of rocks in paleomagnetic research, *Geophys. J., 1*(4), 308–319, 1958.

Brinkmann, R., *Geologic Evolution of Europe,* Hafner Publishing Company, New York, 1960.

Carey, S. W., A tectonic approach to continental drift, *Continental Drift Symposium,* Hobart, Tasmania, pp. 177–355, 1958.

Cox, A., and R. Doell, Review of paleomagnetism, *Bull. Geol. Soc. Am., 71,* 645–768, 1960.

Creer, K. M., E. Irving, and A. E. M. Nairn, Paleomagnetism of the Great Whin Sill, *Geophys. J., 2*(4), 306–323, 1959.

Creer, K. M., E. Irving, and S. K. Runcorn, The direction of the geomagnetic field in remote epochs in Great Britain, *J. Geomagnet. Geoelec., 6,* 163–168, 1954.

De Boer, J., The geology of the Vicentinian Alps (NE-Italy) (with special reference to their paleomagnetic history), *Geol. Ultraiectina, 11,* 1963.

De Sitter, L. U., *Structural Geology,* McGraw-Hill Book Company, New York, 1956.

Dietzel, G. F. L., Geology and Permian paleomagnetism of the Merano region, Province of Bolzano (N. Italy), *Geol. Ultraiectina, 4,* 1960.

Du Bois, P. M., Comparison of paleomagnetic results for selected rocks of Great Britain and North America, *Advan. Phys., 6,* 177–186, 1957.

Findhammer, Unpublished report of fieldwork in N. Italy, Geological Institute of Utrecht, Holland.

Furon, R., *The Geology of Africa,* Haffner Publishing Company, New York, 1963.

Glangeaud, L., and C. Bobier, Corrélations entre le paléomagnetisme et les phases orogéniques pendant le Tertiaire, *Compt. Rend. Soc. Géol. France, 1,* 7–11, 1963.

Guicherit, R., Gravity tectonics, gravity field, and paleomagnetism in NE-Italy (with special reference to the Carnian Alps), *Geol. Ultraiectina, 14,* 1964.

Hodgson, J. H., and J. I. Cock, Direction of faulting in the Greek earthquakes of August 9–13, 1953, *Publ. Dominion Obs. Ottawa, 18*(8), 1956.

Hodgson, J. H., and J. I. Cock, Direction of faulting in some of the larger earthquakes of 1954–1955, *Publ. Dominion Obs. Ottawa, 19*(6), 1957.

Hodgson, J. H., and A. Stevens, Direction of faulting in some of the larger earthquakes of 1955–1956, *Publ. Dominion Obs. Ottawa, 19*(8), 1958.

Hospers, J., Rock magnetism and polar wandering, *Nature, 173*(4416), 1183–1184, 1954.

Kalashnikov, A. C., The history of the geomagnetic field (from paleomagnetic data), *Izv. Akad. Nauk SSSR, Ser. Geofiz.,* 1243–1297, 1961.

Kruseman, G. P., Etude paléomagnetique et sédimentologique du Bassin Permien de Lodève (Hérault), France, *Geol. Ultraiectina, 9,* 1962.

Lookeren Campagne, Unpublished report of fieldwork in N. Italy, Geological Institute of Utrecht, Holland.

Nairn, A. E. M., Observations paléomagnetiques en France: roches Permiennes, *Extr. Bull. Soc. Géol. France, 6*(7), 59–68, 1957.

Nairn, A. E. M., Paleomagnetic results from Europe, *J. Geol., 68*(3), 285–306, 1960.

Nairn, A. E. M., Abstract, Paleoclimate Conference, Newcastle, 1963.

Nairn, A. E. M., Paleomagnetic measurements on Karroo and post Karroo rocks, *Overseas Geol. Min. Resources, G. Brit., 9*(3), 302–320, 1964.

Nijenhuis, G. H. W., Unpublished report of fieldwork in NE-Italy, Geological Institute of Utrecht, Holland.

Nijenhuis, G. H. W., A paleomagnetic study of the Permian volcanics in the Nahe region (SW Germany), *Geol. Mijnbouw, 40,* 26–38, 1960.

Pavoni, N., Die Nordanatolische Horizontalverschiebung, *Geol. Rundschau, 51*(1), 122–139, 1961.

Roche, A., H. Saucier, and J. Lacaze, Etude paléomagnétique des roches volcaniques Permiennes de la région Nideck-Donon, *Bull. Serv. Carte Géol. Alsace Lorraine, 15*(2), 59–68, 1962.

Rod, E., Fault pattern, northwest corner of Sahara shield, *Bull. Am. Assoc. Petrol. Geologists, 46*(4), 529–552, 1962.

Rutten, M. G., et al., Paleomagnetism in the Permian of the Oslo graben (Norway) and of the Esterel (France), *Geol. Mijnbouw, 19e*(6), 1957.

Scheidegger, A. E., *Principles of Geodynamics,* Academic Press, New York, 1963.

Schwarz, E. J., A paleomagnetic investigation of Permo-Triassic red beds and andesites from the Spanish Pyrenees, *J. Geophys. Res., 68*(10), 3265–3271, 1963.

Schwarz, E. J., Geology and paleomagnetism of the Rio Aragon (Spanish Pyrenees), *Estud. Geol. Inst. Invest. Geol. "Lucas Mallada," Madrid,* in press, 1964.

Sougy, J., West African fold belt, *Bull. Geol. Soc. Am., 73*(7), 871–876, 1962.

Swartz, D. H., and D. D. Arden, Geologic history of Red Sea area, *Bull. Am. Assoc. Petrol. Geologists, 44*(10), 1621–1637, 1960.

Van der Lingen, G. J., Geology of the Spanish Pyrenees, north of Canfranc (Huesca), *Estud. Geol. Inst. Invest. Geol. "Lucas Mallada," Madrid, 16,* 205–242, 1960.

Van Everdingen, R. O., Paleomagnetic analyses of Permian extrusives in the Oslo region (Norway), *Det. Norske Videnkap. Akad. 1, Oslo, Mat. Naturv. Kl.,* 1960.

Van Everdingen, R. O., and J. D. A. Zijderveld, Paleomagnetismus in den Rhomben porphyren von Oslo and in den Rhyoliten, Doleriten und Sedimenten des Esterels, *Geol. Rundschau, 48,* 195–205, 1959.

Van Hilten, D., Geology and Permian paleomagnetism of the Val-di-Non area (W. Dolomites, N. Italy), *Geol. Ultraiectina, 5,* 1960.

Van Hilten, D., Presentation of paleomagnetic data, polar wandering and continental drift, *Am. J. Sci., 260,* 401–426, 1962.

Zijderveld, J. D. A., Thesis of Zijderveld with data of Table 2 is in press and will be published in the series *Geol. Ultraiectina,* 1964.

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